

HBN1 — HBN1 TASK 2: DATA ANALYSIS REPORT

DATA ENGINEERING CAPSTONE — D610

PRFA — HBN1

Preparation

Task Overview

Submissions

Evaluation Report

COMPETENCIES

4170.1.2: Implements a Data Analytics Approach

The learner implements a data analytics approach to address a business problem.

INTRODUCTION

Before starting this task, ensure that Task 1 has been approved by a course instructor. If Task 1 has not been approved, your Task 2 submission will be returned without evaluation.

In this task, you will address the real-world research question you identified in Task 1. You will gather relevant data, analyze that data, and recommend a data science solution. This recommendation will be included in a report that you will write, summarizing the key details of your analysis.

Note: You must successfully pass Task 1 before you begin working on Task 2. However, in practice, your topic, data scope, and research question may change based on your findings as you complete your research. Your work on Task 2 may lead you to recognize that more data is needed and that it must be approved data. For this reason, you may need to update your Task 1 forms before Task 2 is complete.

Note: Any information that would be considered confidential, proprietary, or personal in nature should not be included in your actual task submission to WGU. Do not include the actual names of people, suppliers, or companies, or other identifiable information. Fictional names should be used where necessary. Company-specific data, including financial information, should not be included but should be addressed in a general fashion as appropriate.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See [Understanding Similarity Reports](#) for more information.



Grammarly Note:

Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See [Use Grammarly for Education Effectively](#) for more information.

Microsoft Files Note:

Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see [Computer System and Technology Requirements](#).

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

Create a report by doing the following:

Research Question

- A. Summarize the real-data research question you identified in Task 1. Your summary should include justification for the research question you identified in Task 1, a description of the context in which the research question exists, and a discussion of your hypothesis.

Data Collection

- B. Report on your data-collection process by describing the relevant data you collected, discussing **one** advantage and **one** disadvantage of the data-gathering methodology you used, and discussing how you overcame any challenges you encountered during the process of collecting your data.

Data Extraction and Preparation

- C. Describe your data-extraction and -preparation process and provide screenshots to illustrate *each* step. Explain the tools and techniques you used for data extraction and data preparation, including how these tools and techniques were used on the data. Justify why you used these particular tools and techniques, including **one** advantage and **one** disadvantage of using them with your data-extraction and -preparation methods.

Analysis

- D. Report on your data-analysis process by describing the analysis techniques you used to appropriately analyze the data. Include the calculations you performed and their outputs. Justify how you selected the analysis techniques you used, including **one** advantage and **one** disadvantage of these techniques.

Outcomes and Implications

- E. Summarize the implications of your data analysis by discussing the results in the context of the research question, including **one** limitation of your analysis. Within the context of your research question,

recommend a course of action based on your results. Then propose **two** directions or approaches for future study of the dataset.

F. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

G. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A: RESEARCH QUESTION

NOT EVIDENT

A summary of the research question is not provided.

APPROACHING COMPETENCE

The summary of the research question does not include a justification for the question, a description of the context, or a discussion of the hypothesis. 1 or more of these elements do not align with each other, or the hypothesis is implausible given the research question.

COMPETENT

The summary of the research question includes justification for the question, a description of the context, and a discussion of the hypothesis. *Each* of these elements aligns with the others, and the hypothesis is plausible given the research question.

B: DATA COLLECTION

NOT EVIDENT

A report on the data-collection process is not provided.

APPROACHING COMPETENCE

The report on the data-collection process does not include a description of the data collected, a discussion of 1 advantage and 1 disadvantage of the methodology used, or a discussion of how challenges in the process were overcome. 1 or more of these elements do not align with each other, or the

COMPETENT

The report on the data-collection process includes a description of the data collected, a discussion of 1 advantage and 1 disadvantage of the methodology used, and a discussion of how challenges in the process were overcome. *Each* of these elements aligns with each other, and the methodology

methodology used was inappropriate for the data collected.

used was appropriate for the data collected.

C: DATA EXTRACTION AND PREPARATION

NOT EVIDENT

A description of the data-extraction and -preparation process is not provided.

APPROACHING COMPETENCE

The description of the data-extraction and -preparation process does not include screenshots to illustrate *each* step; an explanation of the tools and techniques used, including how they were used on the data; or a justification for the tools and techniques, including 1 advantage or 1 disadvantage. 1 or more of these elements do not align with each other, the methods used were inappropriate for the data, or no evidence of the work is provided.

COMPETENT

The description of the data-extraction and -preparation process includes screenshots to illustrate *each* step; an explanation of the tools and techniques used, including how they were used on the data; and a justification for the tools and techniques, including 1 advantage and 1 disadvantage. *Each* of these elements aligns with the others, the methods used were appropriate for the data, and evidence of the work is provided.

D: ANALYSIS

NOT EVIDENT

A report on the data-analysis process is not provided.

APPROACHING COMPETENCE

The description of the techniques used does not include calculations performed or their outputs. Or the description does not include a justification of why the analysis techniques were selected or 1 advantage or 1 disadvantage of these techniques. Or the description of the techniques used or the analysis is not appropriate for the data."

COMPETENT

The report on the data-analysis process includes a description of the techniques used. The report includes the calculations performed and their outputs as well as a justification of why the analysis techniques were selected and 1 advantage and 1 disadvantage of these techniques. The techniques used and the analysis are appropriate for the data.

E: DATA SUMMARY AND IMPLICATIONS

NOT EVIDENT

APPROACHING COMPETENCE

COMPETENT

A summary of the implications of the data analysis is not provided.

The summary of the implications of the data analysis does not include a discussion of the data-analysis results in the context of the research question or does not include 1 limitation of the analysis. Or the summary does not include a recommended course of action given the research question or 2 directions or approaches for future study of the dataset. Or the recommendation and proposals do not relate to the results of the analysis or could not plausibly be carried out in subsequent research.

The summary of the implications of the data analysis includes a discussion of the data-analysis results in the context of the research question, including 1 limitation of the analysis. The summary also includes a recommended course of action given the research question as well as 2 directions or approaches for future study of the dataset. The recommendation and proposals relate to the results of the analysis and could plausibly be carried out in subsequent research.

F:SOURCES

NOT EVIDENT

The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.

COMPETENT

The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available.

G:PROFESSIONAL COMMUNICATION

NOT EVIDENT

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.